Final TRISO Diffusion and Release Derivation

Supervisor Delivery Package

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1 Scope and assumptions

Five concentric regions are retained: Kernel, Buffer, IPyC, SiC and OPyC, with outer radii $R_1 < \cdots < R_5 = R$. Within each material D_i is constant for the specified nuclide and temperature. Spherical symmetry and ideal concentration/flux continuity are assumed. The primary FV benchmark uses frozen initial kernel inventory, no continuing source after $t = 0$, and an absorbing outer surface. A separate analytical benchmark may use continuous generation and a Robin boundary.

2 Species conservation

$$\frac{d}{dt}\int_V c\,dV = -\int_{\partial V} \mathbf{J}\cdot\mathbf{n}\,dA + \int_V S\,dV. \tag{TRISO-GOV-020}$$

The divergence theorem and arbitrariness of V give

$$\frac{\partial c}{\partial t} + \nabla\cdot\mathbf{J} = S. \tag{TRISO-GOV-030}$$

With Fick's law $\mathbf{J} = -D\nabla c$ (TRISO-GOV-034),

$$\frac{\partial c}{\partial t} = \nabla\cdot(D\nabla c) + S. \tag{TRISO-GOV-040}$$

Spherical symmetry gives

$$\boxed{\frac{\partial c_i}{\partial t} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 D_i \frac{\partial c_i}{\partial r}\right) + S_i.} \tag{TRISO-FV-100}$$

3 Five-layer conditions

At $r = 0$, $\partial c_1/\partial r = 0$. At $r = R_i$,

$$c_i = c_{i+1}, \qquad D_i \partial_r c_i = D_{i+1}\partial_r c_{i+1}.$$

The primary release calculation uses $c_5(R, t) = 0$. A separate Robin benchmark uses $-D_5 c_5'(R) = h[c_5(R) - c_\infty]$. For initial-inventory release,

$$c(r,0) = \begin{cases} c_0, & 0 \le r < R_1, \\ 0, & R_1 < r < R, \end{cases} \qquad S_i = 0 \ (t > 0).$$

4 Analytical/modal reference

For $c = \phi e^{-\Lambda t}$,

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2 D\phi'\right) + \Lambda\phi = 0.$$

With $u = r\phi$, $u'' + k^2 u = 0$, $k^2 = \Lambda/D$, so

$$u_i = A_i \sin(k_i r) + B_i \cos(k_i r), \qquad \phi_i = u_i/r.$$

Centre regularity, interface conditions and the outer condition form $\mathbf{M}(\Lambda)\mathbf{a} = 0$; roots satisfy $\det \mathbf{M}(\Lambda) = 0$. The formal modal solution is

$$c_i = c_{i,\mathrm{ss}} + \sum_{n\geq 1} A_n \phi_{i,n} e^{-\Lambda_n t}.$$

This analytical track is independent; unresolved global five-layer modal completeness/truncation is not used to validate the FV solver.

5 Conservative spherical finite volumes

For cell j,

$$V_j = \frac{4\pi}{3}(r_{j+1/2}^3 - r_{j-1/2}^3), \quad A_{j\pm1/2} = 4\pi r_{j\pm1/2}^2. \tag{TRISO-FV-101}$$

The spherical volume centroid is

$$r_j = \frac{3}{4}\frac{r_{j+1/2}^4 - r_{j-1/2}^4}{r_{j+1/2}^3 - r_{j-1/2}^3}. \tag{TRISO-FV-109A}$$

Integrating the PDE and introducing conservative face conductances gives

$$\boxed{V_j \frac{dc_j}{dt} = G_W(c_{j-1} - c_j) + G_E(c_{j+1} - c_j) + S_j V_j.} \tag{TRISO-FV-155}$$

For a face f with half distances δ_P, δ_E,

$$\boxed{G_f = \frac{A_f}{\delta_P/D_P + \delta_E/D_E}.} \tag{TRISO-FV-140}$$

This series-resistance form handles discontinuous diffusivity and guarantees equal-and-opposite internal face fluxes. At the centre $A = 0$. For the absorbing outer boundary,

$$G_R = \frac{4\pi R^2 D_5}{R - r_N}, \qquad \dot{M}_{\mathrm{rel}} = G_R c_N.$$

6 Backward Euler

$$\frac{V_j}{\Delta t}(c_j^{n+1} - c_j^n) = G_W(c_{j-1}^{n+1} - c_j^{n+1}) + G_E(c_{j+1}^{n+1} - c_j^{n+1}) + S_j^{n+1} V_j.$$

Thus

$$\boxed{-G_W c_{j-1}^{n+1} + \left(\frac{V_j}{\Delta t} + G_W + G_E\right) c_j^{n+1} - G_E c_{j+1}^{n+1} = \frac{V_j}{\Delta t} c_j^n + S_j^{n+1} V_j.} \tag{1}$$

The production solve is tridiagonal. For Forward Euler verification only,

$$\boxed{\Delta t \leq \min_j \frac{V_j}{G_W + G_E}.} \tag{TRISO-FV-300}$$

7 Inventory, release and conservation

$$M^n = \sum_j V_j c_j^n, \qquad M_{\mathrm{rel}}^{n+1} = M_{\mathrm{rel}}^n + \Delta t\, G_R c_N^{n+1}.$$

For the no-source absorbing case,

$$M^{n+1} + M_{\mathrm{rel}}^{n+1} = M^0$$

up to solver/floating-point error, and

$$F^n = M_{\mathrm{rel}}^n / M^0, \qquad \dot{F}^n = G_R c_N^n / M^0.$$

8 Verification contract and limitations

Interfaces coincide with FV faces. Adequacy is established by mesh/time refinement, positivity and conservation. Existing reviewed evidence shows approximately second-order spatial convergence on the aligned steady benchmark and first-order Forward-Euler temporal convergence; the production package adds arbitrary per-region meshes and Backward Euler.

Synthetic configurations are verification cases, not literature properties. The original full analytical Rust upload is not committed; the package analytical module is therefore a baseline implementation rather than a verbatim reconstruction. Temperature/fluence dependence, arbitrary source schedules and production Robin release are deferred unless explicitly enabled.